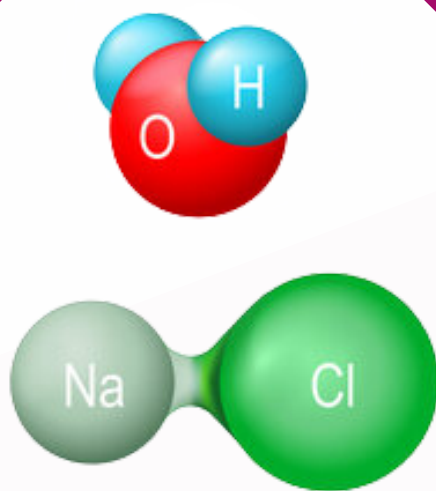


**JEE (MAIN+ADVANCED)**

# Inorganic Chemistry

**ENTHUSIAST COURSE**



**STUDY MATERIAL**

Reactions of Salt Analysis

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English Medium

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Tabulating Reactions of Group-I Cation

| S.No. | Reagents                                                   | Pb <sup>+2</sup>                                                                              | Hg <sub>2</sub> <sup>+2</sup>                                           | Ag <sup>+</sup>                                            |
|-------|------------------------------------------------------------|-----------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|------------------------------------------------------------|
| 1     | (a) Group reagent HCl                                      | PbCl <sub>2</sub> (White)                                                                     | Hg <sub>2</sub> Cl <sub>2</sub> (White)                                 | AgCl (White)                                               |
|       | (b) HCl + NH <sub>3</sub>                                  | PbCl <sub>2</sub> + Pb(OH) <sub>2</sub> (White ppt.)                                          | (Black) Hg + HgNH <sub>2</sub> Cl (White)                               | Dissolve [Ag(NH <sub>3</sub> ) <sub>2</sub> ] <sup>+</sup> |
|       | (c) HCl + boiling                                          | Dissolves                                                                                     | No change                                                               | No change                                                  |
| 2     | (a) KI, small amount                                       | (Yellow) PbI <sub>2</sub>                                                                     | Hg <sub>2</sub> I <sub>2</sub> (Green)                                  | AgI (Yellow)                                               |
|       | (b) KI Excess                                              | No change<br>Conc. I <sup>-</sup> (6M) + PbI <sub>2</sub> → [PbI <sub>4</sub> ] <sup>-2</sup> | Grey Hg + [HgI <sub>4</sub> ] <sup>-2</sup>                             | No change                                                  |
| 3     | K <sub>2</sub> CrO <sub>4</sub>                            | (Yellow) PbCrO <sub>4</sub>                                                                   | (Red) Hg <sub>2</sub> CrO <sub>4</sub>                                  | Brick red Ag <sub>2</sub> CrO <sub>4</sub>                 |
| 4     | (a) KCN                                                    | (White) Pb(CN) <sub>2</sub>                                                                   | (Black) Hg + Hg(CN) <sub>2</sub> soluble                                | (White) AgCN                                               |
|       | (b) Excess KCN                                             | No change                                                                                     | No Change                                                               | [Ag(CN) <sub>2</sub> ] <sup>-</sup> soluble                |
| 5     | (a) NaOH small amount                                      | Pb(OH) <sub>2</sub> (White)                                                                   | Hg <sub>2</sub> O (Black)                                               | Ag <sub>2</sub> O (Brown)                                  |
|       | (b) Excess of NaOH                                         | Dissolves [Pb(OH) <sub>4</sub> ] <sup>-2</sup>                                                | No change                                                               | No change                                                  |
| 6     | (a) NH <sub>3</sub> small amount                           | Pb(OH) <sub>2</sub> (White)                                                                   | Hg <sub>2</sub> (black) + HgO.HgNH <sub>2</sub> NO <sub>3</sub> (White) | Ag <sub>2</sub> O (Brown)                                  |
|       | (b) Excess NH <sub>3</sub>                                 | No Change                                                                                     | No Change                                                               | [Ag(NH <sub>3</sub> ) <sub>2</sub> ] <sup>+</sup> soluble  |
| 7     | H <sub>2</sub> S in acidic medium (H <sub>2</sub> S + HCl) | PbS (Black)                                                                                   | Black Hg + HgS                                                          | Black Ag <sub>2</sub> S                                    |
| 8     | (a) Na <sub>2</sub> CO <sub>3</sub>                        | White ppt of a mixture of Pb(OH) <sub>2</sub> + PbCO <sub>3</sub> or PbO.PbCO <sub>3</sub>    | Hg <sub>2</sub> CO <sub>3</sub> yellowish white                         | Ag <sub>2</sub> CO <sub>3</sub> yellowish white            |
|       | (b) Na <sub>2</sub> CO <sub>3</sub> + Boiling              | No change                                                                                     | (Black)Hg + HgO ↓ (Yellow)<br>Blackish-grey                             | Brown Ag <sub>2</sub> O                                    |

|           |                                       |                                                                                                                                                            |                                                             |                                                                                                                                                                                                        |
|-----------|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>9</b>  | $\text{Na}_2\text{HPO}_4$             | (White) $\text{Pb}_3(\text{PO}_4)_2$                                                                                                                       | (White) $\text{Hg}_2\text{HPO}_4$                           | (Yellow) $\text{Ag}_3\text{PO}_4$                                                                                                                                                                      |
| <b>10</b> | (a) $\text{Na}_2\text{SO}_3$          | $\text{PbSO}_3$ (White)                                                                                                                                    | -                                                           | No visible charge occurs due to formation of complex initially but the addition of more reagent white $\text{Ag}_2\text{SO}_3$ is formed                                                               |
|           | (b) Boiling                           | ppt. oxidized by atmospheric oxygen and $\text{PbSO}_4$ is formed                                                                                          | -                                                           | Grey metallic silver is precipitated                                                                                                                                                                   |
| <b>11</b> | (a) $\text{Na}_2\text{S}_2\text{O}_3$ | First there is no change but further addition of the reagent a white ppt. of $\text{PbS}_2\text{O}_3$ is formed                                            | -                                                           | At first no ppt occurs because the soluble complex $[\text{Ag}(\text{S}_2\text{O}_3)_2]^{-3}$ is formed but further addition of the reagent a white ppt of $\text{Ag}_2\text{S}_2\text{O}_3$ is formed |
|           | (b) Boiling                           | On boiling the suspension the precipitate darkens forming finally black ppt of $\text{PbS} \downarrow$                                                     | -                                                           | The ppt. is unstable, turning dark on standing, when $\text{Ag}_2\text{S}$ is formed this hydrolytic decomposition can be accelerated by warming                                                       |
| <b>12</b> | Specific reactions                    | (i) With $\text{K}_2\text{CrO}_4 \downarrow$<br>yellow $\text{PbCrO}_4 \downarrow$<br>(ii) with Diphenylthiocarbazone gives brick red complex $\downarrow$ | With Di phenyl carbazide<br>$\downarrow$<br>(violet colour) | With p-dimethylamino benzylidenerhodanine<br>(+HNO <sub>3</sub> ) $\rightarrow$ (violet colour).                                                                                                       |

Tabulating Reactions of Group-II Cation

| S.No. | Reagents                                                  | Pb <sup>+2</sup>                                                                         | Cu <sup>+2</sup>                                                           | Hg <sup>+2</sup>                                                                                           | Cd <sup>+2</sup>                                                | Bi <sup>+3</sup>                                                                                                                                                                                                                                                                                       |
|-------|-----------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1     | Group reagent<br>H <sub>2</sub> S gas in<br>acidic medium | PbS (Black)↓                                                                             | CuS (Black) ↓                                                              | HgS (Black) ↓                                                                                              | CdS<br>(Yellow)↓                                                | Bi <sub>2</sub> S <sub>3</sub><br>(Brownish Black)↓                                                                                                                                                                                                                                                    |
| 2     | (a) KI (Little)                                           | PbI <sub>2</sub> ↓ (deep yellow)                                                         | CuI(White) ↓ + I <sub>3</sub> <sup>-</sup> (Brown)<br>Brown colouration    | Hg <sub>2</sub> (Scarlet Red)                                                                              | No reaction                                                     | BiI <sub>3</sub> (Black)↓                                                                                                                                                                                                                                                                              |
|       | (b) Excess KI                                             | PbI <sub>2(s)</sub> + I <sup>-</sup> (6M) →<br>[PbI <sub>4</sub> ] <sup>-2</sup> soluble | No reaction                                                                | [HgI <sub>4</sub> ] <sup>-2</sup><br>Soluble (colourless)                                                  | –                                                               | BiI <sub>3</sub> + I <sup>-</sup> ⇌ [BiI <sub>4</sub> ] <sup>-</sup><br>Orange solution when<br>diluted with water the<br>above reaction is<br>reversed and black<br>BiI <sub>3</sub> is reprecipitated<br>BiI <sub>3</sub> $\xrightarrow[\text{H}_2\text{O}]{\text{Boil}}$ BiOI<br>(orange turbidity) |
| 3     | (a) KCN (Little)                                          | Pb(CN) <sub>2</sub> (white ppt.)↓                                                        | Cu(CN) <sub>2</sub> (Yellow) →<br>CuCN↓(White) + (CN) <sub>2</sub> ↑       | Hg(CN) <sub>2</sub> (soluble)                                                                              | Cd(CN) <sub>2</sub><br>(White ppt.)                             | Bi(CN) <sub>3</sub> (White ppt.)                                                                                                                                                                                                                                                                       |
|       | (b) Excess KCN                                            | No reaction                                                                              | K <sub>3</sub> [Cu(CN) <sub>4</sub> ]<br>colourless solution               | –                                                                                                          | K <sub>2</sub> [Cd(CN) <sub>4</sub> ]<br>colourless<br>solution | No reaction                                                                                                                                                                                                                                                                                            |
| 4     | KSCN                                                      | Pb(SCN) <sub>2</sub> ↓ (White)                                                           | Cu(SCN) <sub>2</sub> ↓ (Black) →<br>CuSCN ↓ +(SCN) <sub>2</sub><br>(White) | Co <sup>+2</sup> +4SCN <sup>-</sup> +Hg <sup>+2</sup> →<br>Co[Hg(SCN) <sub>4</sub> ] ↓<br>(Deep Blue ppt.) | No reaction                                                     | Bi(SCN) <sub>3</sub> (White)                                                                                                                                                                                                                                                                           |
| 5     | (a) NaOH                                                  | Pb(OH) <sub>2</sub> (White<br>ppt.↓)                                                     | Cu(OH) <sub>2</sub><br>(Light blue ppt.↓)                                  | HgO (Yellow ppt.)                                                                                          | Cd(OH) <sub>2</sub><br>(White ppt.)                             | Bi(OH) <sub>3</sub><br>(White ppt.)                                                                                                                                                                                                                                                                    |
|       | (b) Excess<br>NaOH                                        | [Pb(OH) <sub>4</sub> ] <sup>-2</sup> soluble                                             | Insoluble<br>(No change)                                                   | Insoluble<br>(No change)                                                                                   | Insoluble<br>(No change)                                        | Insoluble<br>(No change)                                                                                                                                                                                                                                                                               |
| 6     | SnCl <sub>2</sub>                                         | PbCl <sub>2</sub> (White)                                                                | CuCl (White)                                                               | Hg <sub>2</sub> Cl <sub>2</sub> (White)<br>if excess SnCl <sub>2</sub> then<br>Hg (grey-black)             | (No<br>reaction)                                                | Bi (Black) in basic<br>medium                                                                                                                                                                                                                                                                          |

Tabulating Reactions of Group-III Cation

| S.No. | Reagents                      | Al <sup>3+</sup>                                    | Cr <sup>3+</sup>                                                                                        | Fe <sup>3+</sup>                                                                                                | Fe <sup>+2</sup>                                              |
|-------|-------------------------------|-----------------------------------------------------|---------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|
| 1     | KI                            | —                                                   | —                                                                                                       | $\text{Fe}^{+3} + \text{I}^- \rightarrow \text{Fe}^{+2} + \text{I}_2$ (redox)                                   | $\text{Fe}^{+2} + \text{I}^- \rightarrow \text{FeI}_2$        |
| 2     | (i) KCN                       | —                                                   | —                                                                                                       | $\text{Fe}(\text{CN})_3$ (Brown ppt.)                                                                           | $\text{Fe}(\text{CN})_2$<br>(Yellowish-brown)                 |
|       | (ii) Excess KCN               | —                                                   | —                                                                                                       | $\text{K}_3[\text{Fe}(\text{CN})_6]$<br>(Yellow solution)                                                       | $\text{K}_4[\text{Fe}(\text{CN})_6]$<br>(Pale-yellow sol.)    |
| 3     | KSCN                          | —                                                   | —                                                                                                       | $[\text{Fe}(\text{H}_2\text{O})_5(\text{SCN})]^{+2}$<br>Blood red colouration                                   | No colouration                                                |
| 4     | (a) $(\text{NH}_4)_2\text{S}$ | $\text{Al}(\text{OH})_3$<br>(White gelatinous ppt.) | $\text{Cr}(\text{OH})_3$ (Green ppt.)                                                                   | $\text{FeS} \downarrow + \text{S} \downarrow$<br>Black      White<br>Black ppt.                                 | FeS (Black)                                                   |
|       | (b) If add HCl                | $\text{Al}(\text{OH})_3$ dissolve                   | $\text{Cr}(\text{OH})_3$ dissolve                                                                       | Turns white because<br>FeS is dissolve<br>$\text{FeCl}_2 + \text{H}_2\text{S} \uparrow + \text{S}$ (White ppt.) | FeS dissolve<br>$\text{FeCl}_2 + \text{H}_2\text{S} \uparrow$ |
| 5     | (a) NaOH (Little)             | $\text{Al}(\text{OH})_3$ (White)                    | $\text{Cr}(\text{OH})_3$ (Green)                                                                        | $\text{Fe}(\text{OH})_3$<br>(Reddish-brown)                                                                     | $\text{Fe}(\text{OH})_2$ (Dirty Green)                        |
|       | (b) Excess NaOH               | Dissolves $[\text{Al}(\text{OH})_4]^-$              | Dissolves<br>$[\text{Cr}(\text{OH})_4]^-$ (Green)                                                       | Insoluble                                                                                                       | Insoluble                                                     |
| 6     | (a) $\text{NH}_3$ (Little)    | $\text{Al}(\text{OH})_3$ (White)                    | $\text{Cr}(\text{OH})_3$ (Green ppt.)                                                                   | $\text{Fe}(\text{OH})_3$<br>(Reddish-brown)                                                                     | $\text{Fe}(\text{OH})_2$ (Dirty Green)                        |
|       | (b) Excess $\text{NH}_3$      | Insoluble                                           | Slightly soluble due<br>to formation of<br>$[\text{Cr}(\text{NH}_3)_6]^{+3}$ violet or<br>pink solution | No reaction                                                                                                     | No reaction                                                   |

|    |                 |                                                                                              |                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                   |                                                                                                               |
|----|-----------------|----------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| 7  | $K_4[Fe(CN)_6]$ | –                                                                                            | –                                                                                                                                                                                                                                                                                                                 | $Fe_4[Fe(CN)_6]_3$ (Prussian Blue)<br>soluble in oxalic acid                                                                                      | $K_2Fe[Fe(CN)_6]$ (White)                                                                                     |
| 8  | $K_3[Fe(CN)_6]$ | –                                                                                            | –                                                                                                                                                                                                                                                                                                                 | $Fe[Fe(CN)_6]$ (Brown)                                                                                                                            | $Fe_3[Fe(CN)_6]_2$<br>(Turnbull's Blue)                                                                       |
| 9  | Specific test   | Add $CH_3COONa$<br>no change but boil<br>$Al(OH)_3 \cdot CH_3COO \downarrow$<br>(White ppt.) | Add 1, 5-Diphenylcarbazide<br>$\downarrow$<br>(Violet colour)<br>(a) With $Na_2O_2 \rightarrow$<br>$CrO_4^{2-}$ (Yellow)<br>(b) With $BO_3^- + OH^- \rightarrow$<br>$CrO_4^{2-}$ (Yellow)<br>(c) With $OBr^- \rightarrow$<br>$CrO_4^{2-}$ (Yellow)<br>(d) With $S_2O_8^{2-} \rightarrow$<br>$CrO_4^{2-}$ (Yellow) | Add $CH_3COONa \rightarrow$<br>Deep red colour complex<br>$[Fe(OH)_2(CH_3COO)_6]^+$<br>On boiling $Fe(OH)_2 \cdot CH_3COO$<br>(Brownish Red ppt.) | Add dipy or phen. red colour<br>Add dmg (ammonical)<br>Red colouration ( $Ni^{+2}$ , $Co^{+2}$ )<br>interfere |
| 10 | $Na_2CO_3$      | $Al(OH)_3$ (White)                                                                           | $Cr(OH)_3$ (Green)                                                                                                                                                                                                                                                                                                | $Fe(OH)_3$ (Reddish-brown)                                                                                                                        | –                                                                                                             |

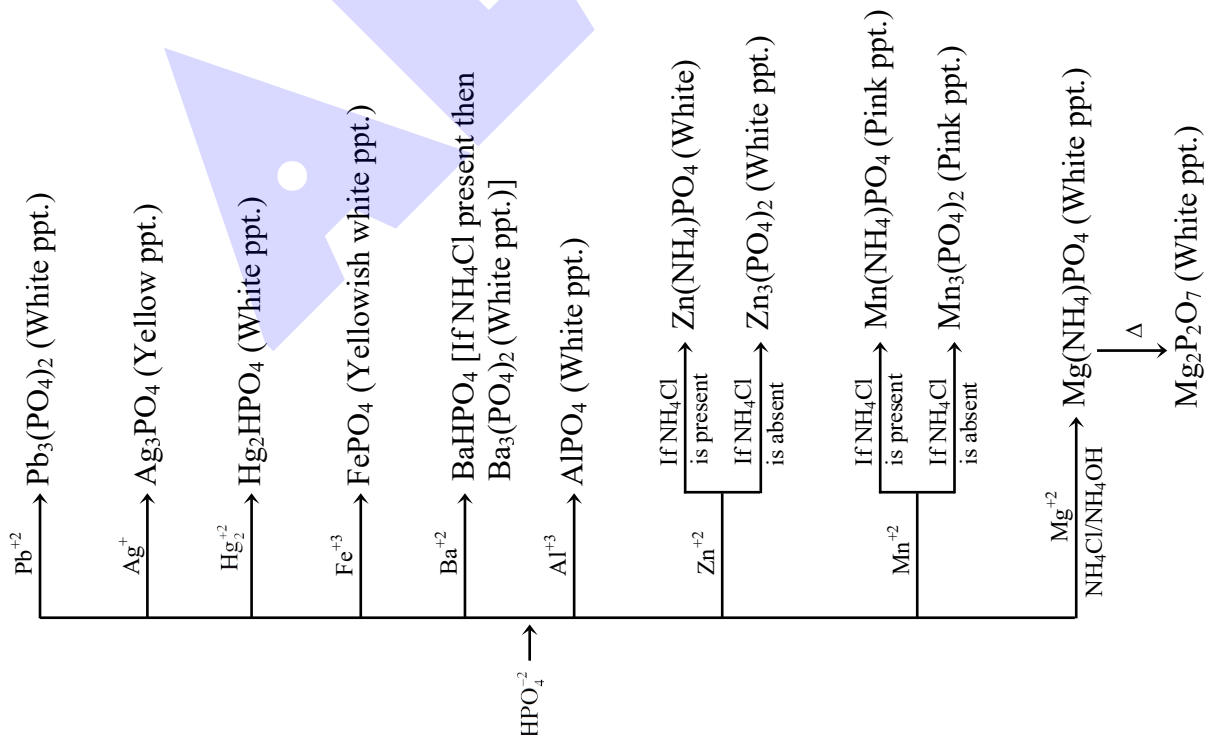
Tabulating Reactions of Group-IV Cation

| S.No. | Reagents                              | Ni <sup>+2</sup>                                                                                                                                                  | Co <sup>+2</sup>                                                                                                                                                                                                                                                            | Mn <sup>+2</sup>                                | Zn <sup>+2</sup>                                                                                                  |
|-------|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| 1     | (i) (NH <sub>4</sub> ) <sub>2</sub> S | NiS (Black)                                                                                                                                                       | CoS (Black)                                                                                                                                                                                                                                                                 | MnS (Pink or Buff coloured)                     | ZnS (White)                                                                                                       |
|       | (ii) Add HCl in sulphide ppt.         | No change (requires aqua regia)                                                                                                                                   | No Change (requires aqua regia)                                                                                                                                                                                                                                             | Dissolves, Mn <sup>+2</sup> + H <sub>2</sub> S↑ | dissolves Zn <sup>+2</sup> + H <sub>2</sub> S↑                                                                    |
| 2     | H <sub>2</sub> S (No reaction)        | NiS (Partial ppt.)                                                                                                                                                | No reaction                                                                                                                                                                                                                                                                 | No reaction                                     | Partial precipitation of Zns in neutral solution. When conc. of acid (HCl) is about 0.3 M [pH about 0.6], no ppt. |
| 3     | (i) Aq. NaOH (Little)                 | Ni(OH) <sub>2</sub> (Green)                                                                                                                                       | Co(OH)Cl (Blue)                                                                                                                                                                                                                                                             | Mn(OH) <sub>2</sub> (Pink ppt.)                 | Zn(OH) <sub>2</sub> (White)                                                                                       |
|       | (ii) Excess NaOH                      | Insoluble                                                                                                                                                         | Co(OH) <sub>2</sub> (Pink ppt.)                                                                                                                                                                                                                                             | Insoluble                                       | [Zn(OH) <sub>4</sub> ] <sup>2-</sup> or Na <sub>2</sub> ZnO <sub>2</sub> Soluble colourless                       |
| 4     | (i) NH <sub>3</sub> (Little)          | Ni(OH) <sub>2</sub> (Green)                                                                                                                                       | Co(OH)Cl (Blue)                                                                                                                                                                                                                                                             | Mn(OH) <sub>2</sub> (Pink ppt.)                 | Zn(OH) <sub>2</sub> (White)                                                                                       |
|       | (i) Excess NH <sub>3</sub>            | [Ni(NH <sub>3</sub> ) <sub>4</sub> ] <sup>+2</sup> (Blue solution)                                                                                                | [Co(NH <sub>3</sub> ) <sub>6</sub> ] <sup>+2</sup> (Yellow)                                                                                                                                                                                                                 | Insoluble                                       | [Zn(NH <sub>3</sub> ) <sub>4</sub> ] <sup>+2</sup> Colourless solution                                            |
| 5     | KCN (Little)                          | Ni(CN) <sub>2</sub> (Green ppt.)                                                                                                                                  | Co(CN) <sub>2</sub> (Reddish-brown)                                                                                                                                                                                                                                         | Mn(CN) <sub>2</sub> .Mn(OH) <sub>2</sub> Buff   | Zn(CN) <sub>2</sub> (White)                                                                                       |
| 6     | Excess KCN                            | [Ni(CN) <sub>4</sub> ] <sup>-2</sup> (Soluble)                                                                                                                    | [Co(CN) <sub>6</sub> ] <sup>4-</sup> (Brown solution)                                                                                                                                                                                                                       | —                                               | —                                                                                                                 |
| 7     | K <sub>4</sub> [Fe(CN) <sub>6</sub> ] | Ni <sub>2</sub> [Fe(CN) <sub>6</sub> ]↓ (Light Green)                                                                                                             | Co <sub>2</sub> [Fe(CN) <sub>6</sub> ] (Green)                                                                                                                                                                                                                              | Mn <sub>2</sub> [Fe(CN) <sub>6</sub> ] (White)  | K <sub>2</sub> Zn <sub>3</sub> [Fe(CN) <sub>6</sub> ] <sub>2</sub> (White ppt.)                                   |
| 8     | Specific Test                         | Add. dmg (ammonical)<br>Pink / Rosy Red ppt. of [Ni(dmg) <sub>2</sub> ] similar colouration produce by Fe <sup>+2</sup> , Co <sup>+2</sup> with dmg <sup>-1</sup> | Add.<br>KNO <sub>2</sub> +CH <sub>3</sub> COOH+CO <sub>2</sub> →<br>K <sub>3</sub> [Co(NO <sub>2</sub> ) <sub>6</sub> ]+NO↑+H <sub>2</sub> O<br>Yellow ppt. (Fisher salt)<br>Co <sup>+2</sup> +4SCN <sup>-</sup> →<br>[Co(SCN) <sub>4</sub> ] <sup>-2</sup> (Blue solution) | —<br>—                                          | —<br>—                                                                                                            |

Tabulating Reactions of Group-V Cation

| S.No. | Reagents                                                                                                           | Ba <sup>+2</sup>                                                                                 | Sr <sup>+2</sup>                                                                                   | Ca <sup>+2</sup>                                                                                                  |
|-------|--------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| 1     | (i) (NH <sub>4</sub> ) <sub>2</sub> CO <sub>3</sub><br>in presence of<br>NH <sub>4</sub> Cl and NH <sub>4</sub> OH | BaCO <sub>3</sub> ↓ (White)                                                                      | SrCO <sub>3</sub> ↓ (White)                                                                        | CaCO <sub>3</sub> ↓ (White)                                                                                       |
| 2     | CH <sub>3</sub> COOH                                                                                               | BaCO <sub>3</sub> is soluble                                                                     | SrCO <sub>3</sub> is soluble                                                                       | CaCO <sub>3</sub> is soluble                                                                                      |
| 3     | (NH <sub>4</sub> ) <sub>2</sub> CrO <sub>4</sub>                                                                   | BaCrO <sub>4</sub> (Yellow)                                                                      | SrCrO <sub>4</sub> yellow                                                                          | CaCrO <sub>4</sub> (Yellow) but with conc.<br>(NH <sub>4</sub> ) <sub>2</sub> CrO <sub>4</sub>                    |
| 4     | CH <sub>3</sub> COOH                                                                                               | BaCrO <sub>4</sub> insoluble<br>in dil. CH <sub>3</sub> COOH                                     | SrCrO <sub>4</sub> soluble in CH <sub>3</sub> COOH                                                 | CaCrO <sub>4</sub> soluble in CH <sub>3</sub> COOH                                                                |
| 5     | (NH <sub>4</sub> ) <sub>2</sub> SO <sub>4</sub>                                                                    | BaSO <sub>4</sub> (White)                                                                        | SrSO <sub>4</sub> (White)                                                                          | CaSO <sub>4</sub> (White)                                                                                         |
| 6     | Excess of (NH <sub>4</sub> ) <sub>2</sub> SO <sub>4</sub>                                                          | BaSO <sub>4</sub> is insoluble                                                                   | SrSO <sub>4</sub> insoluble                                                                        | CaSO <sub>4</sub> is soluble due to<br>formation of complex<br>[Ca(SO <sub>4</sub> ) <sub>2</sub> ] <sup>-2</sup> |
| 7     | (NH <sub>4</sub> ) <sub>2</sub> C <sub>2</sub> O <sub>4</sub>                                                      | BaC <sub>2</sub> O <sub>4</sub> (White)<br>readily dissolved by hot dil.<br>CH <sub>3</sub> COOH | SrC <sub>2</sub> O <sub>4</sub> (White)<br>CH <sub>3</sub> COOH does not attack it so<br>insoluble | CaC <sub>2</sub> O <sub>4</sub> (White)<br>insoluble in CH <sub>3</sub> COOH                                      |

### Test of Cations With $\text{Na}_2\text{HPO}_4$ Reagent



### Test of Cations With $\text{K}_4[\text{Fe}(\text{CN})_6]$ Reagent

